

Technical Manual

FastBack®

Models 90E/90E-G2



HEAT AND CONTROL®

Food Processing & Packaging Systems

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The FastBack® Model 90E and all associated product handling equipment is covered by one or more of the following U.S. and foreign patents:

- 5,351,807
- 5,794,757
- 5,699,897
- 5,842,678
- 6,079,548
- 2324074 (U.K.)
- Other patents pending

Proprietary Information

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Disclaimer

The information in this manual is a guide to the safe and efficient operation of equipment supplied by Heat and Control, Inc. only. The manual has been compiled using references from equipment suppliers and from the experience of Heat and Control, Inc.

As most values, intervals, quantities, etc. are somewhat arbitrary, and operating conditions of the machinery vary, this manual is offered as a guide only. It should be used in conjunction with the buyer's own maintenance program, engineering judgment, and safety procedures.

No responsibility, either direct or consequential, can be accepted by Heat and Control, Inc. resulting from the use of this manual.

NOTICES

FastBack® Model 90E Technical Manual

Revision Record for individual Sections of this manual

Starting with Revision 0, the following is a list of current revisions for each section of this manual:

Section	Revision Level for each Section				
	Overall Manual Revision 0	Overall Manual Revision 1	Overall Manual Revision 2	Overall Manual Revision 3	
Front Section 1 - Title Page	0	0	0	2	
Front Section 2 - Notices	0	1	1	3	
Front Section 3 - About This Manual	0	1	1	2	
Front Section 4 - Table of Contents	0	1	1	2	
Front Section 5 - Contact Information	0	1	1	2	
Section C-1 – Safety	0	2	2	3	
Section C-2 – About FastBack® 50E	0	1	1	2	
Section C-3 – Pre-Installation Procedures	0	1	1	2	
Section C-4 – Installation Procedures	0	1	1	2	
Section C-5 – Maintenance Procedures	0	1	1	2	
Appendix A – Troubleshooting	0	1	1	2	
Appendix B – Spare Parts List	0	5	5	6	
Appendix C – Torque Specifications	0	1	2	3	
Appendix D – OEM Component Specifications	0	Not Used	Not Used	Not Used	
Appendix E – Default Inverter Parameter Values	0	Not Used	Not Used	Not Used	
Appendix F – Initializing the Yaskawa Inverter	0	Not Used	Not Used	Not Used	
Appendix F – OEM Documentation	0	0	0	2	
Technical Publication PH001 – Yaskawa Power Module	N/A	4	4	Not Used	
Technical Publication PH002 – Wiring Diagrams	N/A	2	2	3	
Technical Publication PH005 – Ultrasonic Sensor Setup	N/A	0	0	1	
Technical Publication PH006 – PhotoEye Setup	N/A	0	0	0	
Technical Publication PH009 - Yaskawa V1000 VFD	N/A	N/A	N/A	3	
Technical Publication PH010 – Allen-Bradley PowerFlex 70 VFD	N/A	N/A	N/A	0	

Revision Record

The following revisions have been made to the FastBack® Models 50E/90E Technical Manual since the original publication in October 2001:

<u>Revision</u>	<u>Date</u>	<u>By</u>	<u>Description</u>
0	10/23/01	Tom Knodell	- Initial Release
1	09/21/07	Ken Petri	- Major Revisions - Sections 1-5 - Deleted Appendix's D & E - Added TechPub's PH001, PH002, PH005, & PH006 - Converted to include Model 90E
2	3/4/08	Tom Knodell	- Updated Appendix C (Torque Specs)
3	8/02/10	Ken Petri	- Deleted TechPub PH001 - Added TechPub's PH009 & PH010 - Updated Spare Parts list - Revised Section 3 to delete 50E-specific references - Revised all sections to reflect 90E only designations and to exclude any reference to Semipower inverters

About This Manual

The FastBack® Model 90E Technical Manual describes how to install, operate, and maintain the FastBack® Model 90E. This section, About This Manual, describes how the manual is organized and the types of information in each section. Please take the time to read this section before reading further.

Organization

This manual is organized into the following sections:

- **Section 1 – Safety** provides general safety guidelines and emergency shutdown procedures.
- **Section 2 – About the FastBack® Model 90E** describes the subassemblies that make up the FastBack® Models 90E.
- **Section 3 – Pre-Installation Procedures** tells you how to prepare for installing the FastBack® Model 90E.
- **Section 4 – Installation Procedures** describes how to install the FastBack® Models 90E at your facility.
- **Section 5 – Maintenance** describes various procedures for keeping all parts of the FastBack® Model 90E in good working condition.
- **Appendix A – Troubleshooting** provides troubleshooting guidelines.
- **Appendix B – Spare Parts List** lists recommended spare parts for the FastBack® Models 90E, along with recommended quantities of each.
- **Appendix C – Torque Specifications** lists all torque values for bolts and screws used in the FastBack® Model 90E subassemblies.
- **Appendix F – OEM Documentation** contains documentation from the original equipment manufacturers of the components used in the FastBack® Model 90E.
- **Technical Publication PH002 – Wiring Diagrams** contains wiring diagrams for the Heat and Control power module, photoeye, and ultrasonic level sensor.
- **Technical Publication PH005 – Ultrasonic Sensor Setup** contains procedures for positioning and adjusting the ultrasonic level sensor.
- **Technical Publication PH006 – Photoeye Setup** contains procedures for positioning and adjusting the Efector photoeyes.
- **Technical Publication PH009 - Yaskawa V1000 Setup** contains the setup procedure, parameter settings, and wiring diagram for the Yaskawa Model V1000 VFD, the electrical drive package used for FastBack® Model 90E.

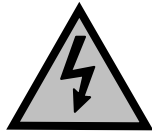
ABOUT THIS MANUAL

FastBack® Model 90E Technical Manual

- **Technical Publication PH010 – Allen-Bradley PowerFlex 70 Setup** contains the setup procedure, parameter settings, and wiring diagram for the Allen-Bradley PowerFlex 70 VFD, the alternative electrical drive package approved for use with FastBack® Model 90E.

Notes for Safe Operation

The following are used throughout this manual.



Indicates an electrical shock hazard. Follow the warnings to avoid death or serious injury.



Indicates a situation dangerous either to you or the equipment. Follow these warnings to avoid serious injury to yourself or damage to the equipment.



Indicates a situation that could cause minor or moderate injury to you or the equipment. Follow these warnings to avoid injury to yourself or damage to the equipment.

Note:

Indicates a step or information necessary to ensure proper operation.

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Tech Pub - PH010	PH010.1

Contact Information

Locations	Headquarters	Service and Training	Manufacturing and Spare Parts	Product Handling Systems Sales and Engineering
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Heat and Control, Ltd. 407 Creek Road Mt. Gravatt Queensland 4122, Australia 011-61-7-3-877-6333 001-61-7-3-343-8371 (fax)		X	X	X
Heat and Control S.A. de C.V. Circuito De La Productividad Sur 107 Parque Industrial Guadalajara El Salto, Jalisco C.P. 45690, Mexico 011-52-33-3689-1146/1147 011-52-33-3689-2240 (fax)			X	X

Table i – Heat and Control, Inc. Locations

SERVICE AND TRAINING

To Report a Problem with the FastBack® Model 90E

If the FastBack® Model 90E fails to operate correctly:

1. Stop the FastBack® by rotating the emergency disconnect switch counterclockwise, to the OFF position.
2. Find the problem, using the troubleshooting guide in Appendix A if necessary.
3. Make a written record of your findings.
4. Call the Heat and Control Service Office in Lititz, PA at (800) 648-5313 or (717) 519-5240. Have the following information ready:
 - FastBack® model and serial number, located on either the power module cover or the underside of the base plate (see Photos i and ii below)
 - Part number of any component needing replacement (see Appendix B for a list of spare parts and part numbers)



Photo i – FastBack® Nameplate on Underside of Base Plate



Photo ii – Closeup of Nameplate

